

PROCESS INTELLIGENCE REPORT

MASTER THESIS SUBMISSION, EVALUATION AND ARCHIVING PROCESS

TAGUS INSTITUTE OF TECHNOLOGY AND MANAGEMENT

A PROJECT BY:
HENRY LEWIS #20222002



TITM
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Executive summary

This report presents a process mining analysis of the master thesis submission, evaluation and archiving process at the Tagus Institute of Technology and Management. The objective was to understand how the process is executed in practice, identify deviations from the expected procedure, evaluate the school director's hypotheses, and provide recommendations that can support operational improvement.

The project was carried out in Celonis using a case table and an event log. The final dashboard analysed **500 thesis cases** and **9,698 recorded events**. The event log covers the process from thesis submission to final notification. The case table provides contextual information such as programme, supervisor, nationality, gender, age and final grade.

The main finding is that the thesis process is highly variable. The dashboard identified **41 process variants**. The most common path covers only **119 cases**, equal to approximately **24%** of the population. The Adherence Explorer showed a **23.8% happy path conformance rate**, which means that most thesis cases deviate from the ideal process in some way or other.

The most important sources of deviation are correction loops, repeated review activity, plagiarism re-checks, final corrected versions, jury revisions, and defense rescheduling. With the evidence suggesting that the process does not mainly suffer from student performance problems. The final grades also appear relatively stable between the groups analyzed. Instead, the strongest explanation is process inconsistency, especially around supervision practices, pre-submission quality control, and scheduling coordination.

The dashboard was structured into **6** business-oriented pages covering variant analysis, process discovery, conformance checking, case exploration, hypothesis testing and an action plan. The final recommendations focus on supervisor development, a pre-submission quality checklist, earlier jury formation, real-time case monitoring and additional scheduling support where rescheduling risk is higher.

Table 1. Executive summary of key findings

Area	Main result
Dataset scope	500 thesis cases and 9,698 events were analysed.
Process variation	41 process variants were identified.
Most common path	The top process path covers 119 cases, around 24% of the population.
Conformance	The Adherence Explorer reports 23.8% happy path conformance.
Main process issue	Correction loops are the largest deviation driver, affecting 216 cases.
Management interpretation	The issue is mainly process consistency and supervision control, not student performance.

1 Introduction

Tagus Institute of Technology and Management is a Portuguese higher education institution located in the Lisbon metropolitan area. Offering programmes in management, finance, informatics and marketing. One of its important academic workflows is the master thesis management process, which includes submission, review, defense, grade validation and final archiving.

The Thesis Management Office coordinates this process across Academic Services, the Library, supervisors and jury members. Although the process is recorded digitally through platforms such as NetPA and internal systems, the school management team raised concerns about efficiency, transparency and consistency. The project brief specifically mentions delays caused by document corrections, issues with structure, references and plagiarism, difficulties in defining or validating jury members, and defense rescheduling due to availability constraints.

Before this project, these issues were mainly based on internal perception. Process mining was therefore used to reconstruct the real process from event data, compare it with the expected procedure and investigate whether the school director's concerns were supported by the data and if these issues stem further than expected.

The central question of the project is:

How is the master thesis management process executed in practice, and which deviations, bottlenecks or contextual factors explain the main process issues?

2 Theoretical background

Process mining is a data-driven method for analysing business processes using event logs. Instead of relying only on interviews or written procedures, process mining reconstructs the process from recorded events. Which makes it useful when the official process and the real process may differ.

A standard event log contains at least three core elements: a case identifier, an activity name and a timestamp. In this project, the case identifier is the thesis case ID, the activity is the step executed in the thesis process, and the timestamp records when the step occurred. The event log also contains the resource responsible for the activity.

Three process mining perspectives were used in this project.

Process discovery reconstructs the actual process flow. The Variant Explorer and Process Explorer were used to understand the most common paths and the amount of variation in the process.

Conformance checking compares the real process against an expected model. The expected model in this project is the 17-step thesis happy path. The Adherence Explorer was used to identify deviations from this expected process.

Performance analysis evaluates process timing and efficiency. Throughput time, activity frequency and the performance spectrum were used to understand how long cases take and where timing differences appear.

The value of process mining is that it links operational evidence to decision-making. It can show not only that a problem exists, but also how many cases are affected, where the problem occurs and which improvement action is most relevant to give targeted solutions.

3 Project description and goals

The project focuses on the end-to-end master thesis process. According to the normal procedure, students submit their thesis by the deadline, but the thesis is only officially accepted after supervisor approval in NetPA and confirmation that all coursework is completed. The Library then reviews structure, references and plagiarism. If issues are found, the thesis is returned for correction and students may resubmit. Once validated, Academic Services requests and validates the jury, schedules the defense and sends invitations. After the defense, grades are submitted in NetPA and validated. If required, students submit a final corrected version. With the process ending with archiving in the institutional repository and a final notification.

The expected happy path used for conformance checking contains 17 main activities.

Table 2. Expected thesis process happy path

Step	Expected activity
1	Receive Thesis Submission
2	Check Course Completion
3	Confirm Final Supervisor Approval
4	Register Submission in NetPA
5	Review Structure and References
6	Check Plagiarism
7	Request Jury Proposal
8	Receive Jury Proposal
9	Validate Jury
10	Schedule Defense
11	Send Defense Invitation
12	Conduct Defense
13	Receive Grades in NetPA
14	Validate Grades
15	Check Need for Final Corrections
16	Archive Thesis in Repository
17	Send Final Notification

The goals of the analysis were to discover the real process flow, identify process variants and rework loops, assess conformance, analyse the role of contextual factors and provide recommendations that can be used by management.

The school director also raised six hypotheses that needed to be evaluated.

Table 3. School director hypotheses

No.	Hypothesis
H1	Certain supervisors are associated with more thesis corrections and rework loops.
H2	Defense rescheduling is mostly due to jury availability and is not significantly influenced by student characteristics.
H3	Informatics students tend to require more final corrections after the defense.
H4	Initial corrections are associated with a higher likelihood of final corrections.
H5	Student performance may differ by demographic groups such as age and gender.
H6	Concentration of issues around supervisors may indicate differences in supervision practices.

4 Data and ETL process

4.1 Input data

The project used two main data sources: a case table and an event log. The case table provides one row per thesis case. The event log records the sequence of activities performed for each case.

Table 4. Case table fields used in the analysis

Field	Use in the analysis
Case ID	Unique thesis case identifier.
Student number	Student identifier.
Nationality	Used to analyse scheduling patterns.
Master programme	Used to compare corrections and performance.
Supervisor	Used to analyse correction and rescheduling concentration.
Gender	Used to compare performance.
Age	Used to explore demographic performance differences.
Final grade	Used as the student performance indicator.

Table 5. Event log fields used in the analysis

Field	Use in the analysis
Case ID	Connects events to each thesis case.
Timestamp	Orders events and supports throughput calculations.
Activity	Identifies each step in the thesis process.
Resource	Identifies the actor responsible for the event.

4.2 Data model and preparation

The case table and event log were loaded into Celonis. The event log was configured as the activity table and connected to the case table using the case ID. This allowed each event to be analysed together with contextual case information such as supervisor, programme, nationality, gender, age and final grade.

The core event log configuration used the case ID as the case identifier, the activity field as the process step, and the timestamp field to order activities in each case. This setup made it possible

to use standard process mining components such as Variant Explorer, Process Explorer, Adherence Explorer, Case Explorer and performance analysis components.

Additional calculated fields and metrics were created in PQL. These included correction loops, rescheduling flags, final correction flags, risk flags, conformance logic and average activity counts per case.

Table 6. ETL and modelling summary

Task	Implementation
Data upload	Loaded the case table and event log into Celonis.
Case connection	Connected the event log to the case table through Case ID.
Timestamp use	Used event timestamps to order activities and calculate throughput.
Activity modelling	Used the activity field to create the process model and conformance analysis.
Context enrichment	Added programme, supervisor, nationality, gender, age and final grade to the analysis.
Calculated metrics	Created correction loop counts, rescheduling indicators, final correction indicators and risk flags.
Dashboard preparation	Renamed technical fields into business-readable labels where possible.

4.3 Main ETL observations

The data was suitable for process mining because the event log contained the required attributes: case ID, timestamp and activity. The case table also provided enough contextual variables to investigate the hypotheses raised by the school director.

There were also limitations. The dataset is offline rather than live, so the dashboard does not represent online process monitoring. Also, the event log records what happened, but it does not always explain why it happened. For example, it records defense rescheduling but does not provide a direct reason for each rescheduling event. Meaning that some interpretations should be treated as process evidence rather than full causal proof and require a bit more investigation.

Table 7. Dataset overview

Item	Value
Cases analysed	500 thesis cases
Events analysed	9,698 events
Activities in dashboard	24 activities
Process variants identified	41 variants
Main case attributes	Programme, supervisor, nationality, gender, age and final grade
Main process questions	Conformance, correction loops, rescheduling, jury revisions and final corrections

5 Dashboard description and main insights

The dashboard was designed to satisfy the required pages with the addition of a final management-oriented page that can be used by less tech-savvy management to get a brief overview at a glance, making the final structure business-readable.

Table 8. Dashboard structure and purpose

Dashboard page	Purpose
Cover	Provides a branded executive entry point with headline KPIs.
Variant Explorer	Shows how fragmented the thesis process is across variants.
Process Explorer	Shows the real process flow, performance spectrum and activity frequency.
Conformance Checking	Compares the real process against the ideal happy path.
Case Explorer and Demographics	Allows case-level drill-down and demographic exploration.
Hypothesis Testing	Answers the six concerns raised by the school director.
Action Plan	Converts findings into monitoring and improvement actions.

5.1 Cover page

The Cover page summarises the project using four headline KPIs. It is designed as an executive entry point rather than a detailed analysis page.

Table 9. Cover page KPIs

KPI	Value shown in dashboard
Theses analysed	500
Average throughput time	102 days
Conformance rate	24%
Corrections	216

The cover page makes the main message clear at the start: the thesis process is large enough to analyse meaningfully, but the conformance rate is low and correction activity is substantial.

5.2 Variant Explorer

The Variant Explorer page shows the degree of process fragmentation. The dashboard identified 41 variants, and the most common variant covers only 119 cases. This means that the most frequent path represents less than one quarter of the full population.

Table 10. Variant Explorer insights

Metric or insight	Result
Total variants	41
Cases in most common variant	119 of 500 cases
Coverage of most common variant	24%
Theses with corrections	216
Average throughput time	102 days
Main interpretation	The official thesis process is not followed consistently by most cases.

The main value of this page is that it demonstrates process instability. If the top path covers only 24% of cases, management cannot rely only on the official procedure to understand the real process. The dashboard therefore helps identify where alternative routes are occurring.

5.3 Process Explorer

The Process Explorer page visualises the actual flow of thesis cases and highlights operational deviations. It combines process discovery with key deviation KPIs and activity frequency.

Table 11. Process Explorer deviation KPIs

Deviation KPI	Value
Rejected submissions	46
Failed defenses	10
Final corrections required	135
Rescheduled defenses	88
Jury revisions	84

This page shows that the process has a clear central path, but there are important loops and side paths around corrections, jury revision and defense scheduling. The Performance Spectrum is useful for understanding the timing of the most common path. Since the selected path covers 119 of 500 cases, it should be interpreted as happy path performance rather than the full process population.

5.4 Conformance Checking

The Conformance Checking page compares the real process with the expected 17-step happy path. The Adherence Explorer reports a 23.8% conformance rate. The dashboard also calculates 119 happy path cases and 381 cases with deviations.

Table 12. Conformance Checking summary

Conformance measure	Value
Happy path cases	119
Cases with deviations	381
Conformance rate	23.8%, displayed as 24%
Cases with corrections	216
Deviation types shown by Adherence Explorer	14

The Adherence Explorer identified the main deviation types. Correction-related deviations dominate, followed by repeated review and plagiarism checks, final corrected versions, rescheduling and jury revisions.

Table 13. Main deviations identified in conformance analysis

Deviation	Affected cases
Request Thesis Corrections	216
Receive Revised Thesis	216
Review Structure and References repeated	144
Receive Final Corrected Version	135
Check Plagiarism repeated	101
Reschedule Defense	88
Request Jury Revision	84

The main conclusion from this page is that non-conformance is not caused by one isolated issue. It is driven by repeated corrections, review loops, plagiarism re-checks and scheduling problems. This supports a process improvement approach rather than a narrow focus on individual student cases.

5.5 Case Explorer and Demographics

The Case Explorer and Demographics page allows users to inspect individual cases and contextual attributes. The table includes case ID, student number, programme, supervisor, nationality, gender, age, final grade and correction loops.

Table 14. Case Explorer analysis areas

Area	Use in the dashboard
Case table	Allows inspection of individual thesis cases.
Average final grade by programme	Compares academic performance across programmes.
Average correction loops per case	Summarises rework intensity at case level.
Gender distribution	Describes the case population.
Nationality distribution	Describes the case population and supports scheduling analysis.
Average final grade by supervisor	Shows whether supervisor issues align with performance differences.

The main insight is that final grades appear relatively stable across the groups analysed. This is important because it suggests that process issues are not mainly explained by weak student performance. The evidence points more strongly toward process control, supervision consistency and administrative coordination.

5.6 Hypothesis Testing

The Hypothesis Testing page investigates the school director’s six hypotheses directly. The final verdicts are summarised in Table 15.

Table 15. Hypothesis testing verdicts

No.	Hypothesis focus	Verdict	Main interpretation
H1	Supervisor correction variation	Confirmed	Correction loops vary strongly by supervisor.
H2	Defense rescheduling drivers	Partially rejected	Reschedules vary across student groups, so jury availability is likely not the only driver.
H3	Informatics correction rate	Rejected	Informatics is not the highest programme for post-defense corrections. Finance shows more post-defense corrections.
H4	Correction loop persistence	Weakly confirmed	Initial corrections slightly increase final correction risk, but the effect is weak.
H5	Demographic performance effect	Rejected	Grades are similar across gender groups and no clear age performance pattern appears.
H6	Supervision practice effect	Confirmed	Issues cluster around certain supervisors while grades remain relatively stable.

The strongest overall conclusion is that process issues are not evenly distributed. Some supervisors and groups show more process friction, but student performance remains relatively stable. This supports the interpretation that the process problem is mainly about supervision consistency and workflow control.

5.7 Action Plan

The Action Plan page translates the analysis into management actions. It includes a summary of key improvement targets, a live case risk monitoring table, an improvement areas chart and a prioritised action panel.

Table 16. Action Plan KPIs

KPI	Value
Corrections	216
Rescheduled defenses	88
Jury revisions	84
Final corrections required	135

The risk monitoring table flags cases using correction loops, defense rescheduling and final correction status. This helps Academic Services prioritise intervention rather than waiting until problems accumulate. It also allows them to export the table so they can analyze it independently.

Table 17. Improvement areas and related process evidence

Improvement area	Evidence used in the dashboard
Pre-defense corrections	Request Thesis Corrections, affecting 216 cases.
Post-defense corrections	Receive Final Corrected Version, affecting 135 cases.
Defense scheduling	Reschedule Defense, affecting 88 cases.
Jury formation	Request Jury Revision, affecting 84 cases.

6 Answer to the school director

6.1 H1: Supervisor correction variation

Verdict: Confirmed.

The analysis shows that correction loops vary strongly by supervisor. Some supervisors are associated with higher average correction-loop rates than others. This suggests that supervision practices may influence the amount of rework required before a thesis can progress.

The result should not be interpreted as a personal criticism of supervisors. It should be used as a management signal that supervision processes may not be standardised. Some students may receive clearer early guidance than others, leading to different levels of correction activity later in the process.

6.2 H2: Defense rescheduling drivers

Verdict: Partially rejected.

Jury availability is likely an important part of defense rescheduling, but the dashboard suggests that it is not the only explanation. Rescheduling varies across student groups. This indicates that other factors, such as coordination, availability windows, communication and administrative planning, may also contribute.

This conclusion should be interpreted carefully. A pattern by nationality or other student characteristic does not prove causality. It shows that rescheduling is not evenly distributed and should be investigated with additional operational information.

6.3 H3: Informatics post-defense correction rate

Verdict: Rejected.

The analysis does not support the hypothesis that Informatics students require the most post-defense corrections. The dashboard shows Finance as the programme with more post-defense corrections than Informatics.

This finding is important because it challenges a prior perception held by management. Improvement work should not focus only on Informatics. Post-defense correction procedures should be reviewed across programmes, with particular attention to Finance.

6.4 H4: Correction loop persistence

Verdict: Weakly confirmed.

Cases with initial corrections show a slightly higher final correction risk than cases without initial corrections. This suggests that early quality issues can persist into the final stage, but the relationship is not strong enough to treat initial correction status as a reliable predictor on its own.

The practical use of this finding is as an early warning signal. Cases with initial correction loops should receive closer monitoring, especially when correction activity happens more than once.

6.5 H5: Demographic performance effect

Verdict: Rejected.

The analysis does not show meaningful performance differences across gender groups. The dashboard also does not indicate a clear age performance pattern. This means that demographic performance differences are not a convincing explanation for the observed process problems.

The implication is that management should avoid framing the problem as student performance or demographic performance. The evidence points more strongly toward process execution and process control.

6.6 H6: Supervision practice effect

Verdict: Confirmed.

The analysis supports the idea that issues cluster around certain supervisors. Correction loops and rescheduling patterns are not evenly distributed. At the same time, final grades remain relatively stable, which suggests that process friction is not simply caused by weaker students.

This is one of the most important findings. It suggests that TITM can create value by standardising supervision practices, clarifying review expectations and identifying cases that need support earlier.

7 Final recommendations and value generated

The dashboard generates value by turning event log data into operational decisions. Rather than only showing that problems exist, it identifies the scale of each issue, where it occurs and which management action is most relevant.

Table 18. Prioritised recommendations

Priority	Recommendation	Expected value
1	Targeted supervisor development	Reduce repeated rework by supporting supervisors with higher correction-loop rates.
2	Pre-submission quality checklist	Reduce avoidable structure, reference and plagiarism corrections before formal review.
3	Earlier jury formation	Reduce jury revisions and defense rescheduling pressure.
4	Real-time risk monitoring	Flag cases with correction loops, reschedules or final corrections for earlier intervention.
5	International scheduling support	Provide extra scheduling support where reschedule rates are higher.

7.1 Recommendation 1: Targeted supervisor development

Correction loops vary by supervisor. TITM should identify supervisors with consistently high correction-loop rates and provide support, guidance or standardised review criteria. This should be presented as a process quality initiative rather than as a penalty.

7.2 Recommendation 2: Pre-submission quality checklist

Correction loops are the largest source of process deviation, affecting 216 cases. A mandatory pre-submission checklist should be introduced before the thesis enters formal review. The checklist should cover structure, references, formatting and plagiarism readiness.

7.3 Recommendation 3: Earlier jury formation

Defense rescheduling and jury revision affect a meaningful number of cases. TITM should begin jury formation earlier to reduce pressure in October and November. Earlier planning would give Academic Services more time to solve availability conflicts before defense scheduling becomes urgent.

7.4 Recommendation 4: Real-time risk monitoring

The Action Plan page includes a risk monitoring table that flags cases with correction loops, defense rescheduling or final corrections. This should be used regularly by Academic Services to identify cases that need intervention.

7.5 Recommendation 5: International scheduling support

Rescheduling patterns vary across student groups. TITM should provide additional scheduling support where rescheduling rates are higher. This does not mean that student characteristics cause rescheduling, but it does show where coordination support may be useful.

8 Limitations

The project provides useful insights, but several limitations should be considered.

First, the dataset is offline. This means that the dashboard is not connected to live operational data. So a real-time implementation would require updated event feeds and operational integration but provide the most accurate results.

Second, the event log records what happened, but not always why it happened. For example, a defense rescheduling event is visible, but the exact reason is not always recorded, limiting causal interpretation.

Third, the analysis identifies associations rather than proving causality. Higher correction-loop rates for a supervisor do not automatically prove that the supervisor caused the corrections. The result shows a pattern that should be investigated further.

Fourth, some demographic groups are smaller than others. Average-based metrics reduce raw-count bias, but results for smaller groups may still be more volatile.

Finally, conformance results depend on the model definition. The Adherence Explorer conformance rate is the preferred measure because it compares the real process against the expected process model. Manual calculations can differ if they include only selected deviations.

Table 19. Main limitations and implications

Limitation	Implication
Offline dataset	The dashboard supports retrospective analysis rather than live monitoring.
Missing reason codes	Some causes, such as why defenses are rescheduled, cannot be proven directly.
Association not causation	Patterns by supervisor or nationality need further investigation before policy changes.
Small group sizes	Some demographic comparisons may be more volatile.
Model-dependent conformance	The definition of the ideal process affects conformance results.

9 Conclusion

To conclude: This project used process mining to analyse the master thesis submission, evaluation and archiving process at TITM. The analysis showed that the process is highly variable, with 41 process variants across 500 cases and only 23.8% happy path conformance.

The main issues are correction loops, repeated review and plagiarism checks, post-defense corrections, jury revisions and defense rescheduling. These issues create additional workload for Academic Services, the Library, supervisors and students. They also make the process harder to predict and manage.

The school director's hypotheses were clarified by the data. Supervisor correction variation was confirmed. Informatics was not the highest programme for post-defense corrections. Demographic performance differences were not supported. The evidence points most strongly toward process consistency and supervision practice as the key improvement areas.

The final dashboard provides a user-friendly way to explore the as-is process, understand deviations, investigate case-level details and monitor at-risk thesis cases. The Action Plan page translates the analysis into management actions. The most important recommendation is to reduce avoidable process variation through supervisor development, pre-submission quality checks, earlier jury formation and live case monitoring.

Overall, the project shows that process mining can help TITM move from subjective perceptions of process problems to evidence-based process improvement and hopefully allow them to streamline their processes to save time and effort down the line.